

WEEK 5 POWER BI REPORT

Healthcare Data Transformation, Data Modeling and Measures

Submitted By

Harini R – E25DA021

1. Introduction

This Week 5 practical work focuses on preparing a healthcare dataset in Power BI, creating dimension and fact tables, merging related data, establishing relationships between tables, and creating measures for analysis. The screenshots show the use of Power Query and the Power BI Model view to organize the data into a structured analytical model.

2. Objectives

- To clean and organize the healthcare dataset using Power Query.
- To create separate patient, admission and billing tables for structured analysis.
- To merge related tables using matching columns.
- To establish relationships between fact and dimension tables.
- To create a dedicated Measures Table for reusable calculations.
- To prepare the model for further healthcare data analysis and reporting.

3. Data Preparation in Power Query

The healthcare dataset contains fields such as Name, Age, Gender, Blood Type, Medical Condition,

Admission Type, Doctor, Hospital, Medication, Room Number, Billing Amount, Date of Admission, Discharge Date and Test Results. The screenshots show valid data profiling and transformation steps. A patient dimension table was prepared by removing unnecessary columns, removing duplicates, adding an index and renaming the index as Patient ID.

Figure 1. Healthcare dataset and data-quality view

Patient Name	Age	Gender	Blood Type	Medical Condition
Jobby Jackson	30	Male	B-	Cancer
eslie Terry	62	Male	A+	Obesity
Janny Smith	76	Female	A-	Obesity
Andrew Watts	28	Female	O+	Diabetes
Adrienne Bell	43	Female	AB+	Cancer
Emily Johnson	36	Male	A+	Asthma
Edward Edwards	21	Female	AB-	Diabetes
Christina Martinez	20	Female	A+	Cancer
Asmine Aguilar	82	Male	AB+	Asthma
Christopher Berg	58	Female	AB-	Cancer
Michelle Daniels	72	Male	O+	Cancer
Yaron Martinez	38	Female	A-	Hypertension
Donnor Hansen	75	Female	A+	Diabetes
Robert Bauer	68	Female	AB+	Asthma

4. Merging Related Tables

The Merge operation was used in Power Query to combine information from the Billing table and the Dim_Admission table. The matching columns were selected so that related admission information could be connected with billing records. The join kind shown in the screenshot is Left Outer, which keeps all records from the first table and brings matching records from the second table.

Figure 2. Merge operation between Billing and Dim_Admission

Merge

Select a table and matching columns to create a merged table.

Billing

Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results	Age Group
18856.28131	328	Urgent	2/2/2024	Paracetamol	Normal	19 - 35
33643.32729	265	Emergency	8/26/2019	Ibuprofen	Inconclusive	Above
27955.09608	205	Emergency	10/7/2022	Aspirin	Normal	Above
37909.78241	450	Elective	12/18/2020	Ibuprofen	Abnormal	19 - 35

Dim_Admission

Admission Type	Doctor	Hospital	Medical Condition	Medication	Room Number
Urgent	Matthew Smith	Sons and Miller	Cancer	Paracetamol	
Emergency	Samantha Davies	Kim Inc	Obesity	Ibuprofen	
Emergency	Tiffany Mitchell	Cook PLC	Obesity	Aspirin	
Elective	Kevin Wells	Hernandez Rogers and Vang,	Diabetes	Ibuprofen	

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

5. Data Model Design

The model view shows a structured arrangement containing Dim_Patient, Fact_Billing and Dim_Admission.

The Fact_Billing table contains measures such as Billing Amount and fields including Admission ID, Date of Admission, Discharge Date, Insurance Provider and Patient ID. The dimension tables provide descriptive information used to filter and analyze the fact data.

Relationships are established between the patient dimension and billing data, and between admission information and billing data. This structure supports organized filtering and aggregation in Power BI.

Figure 3. Power BI model view

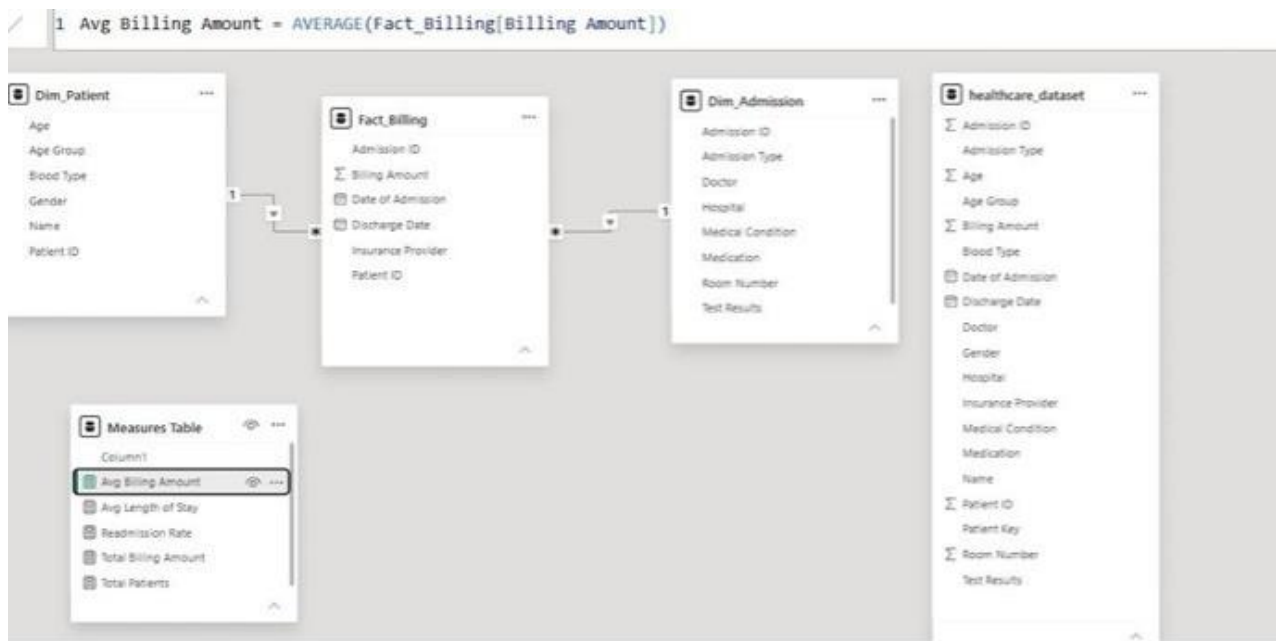
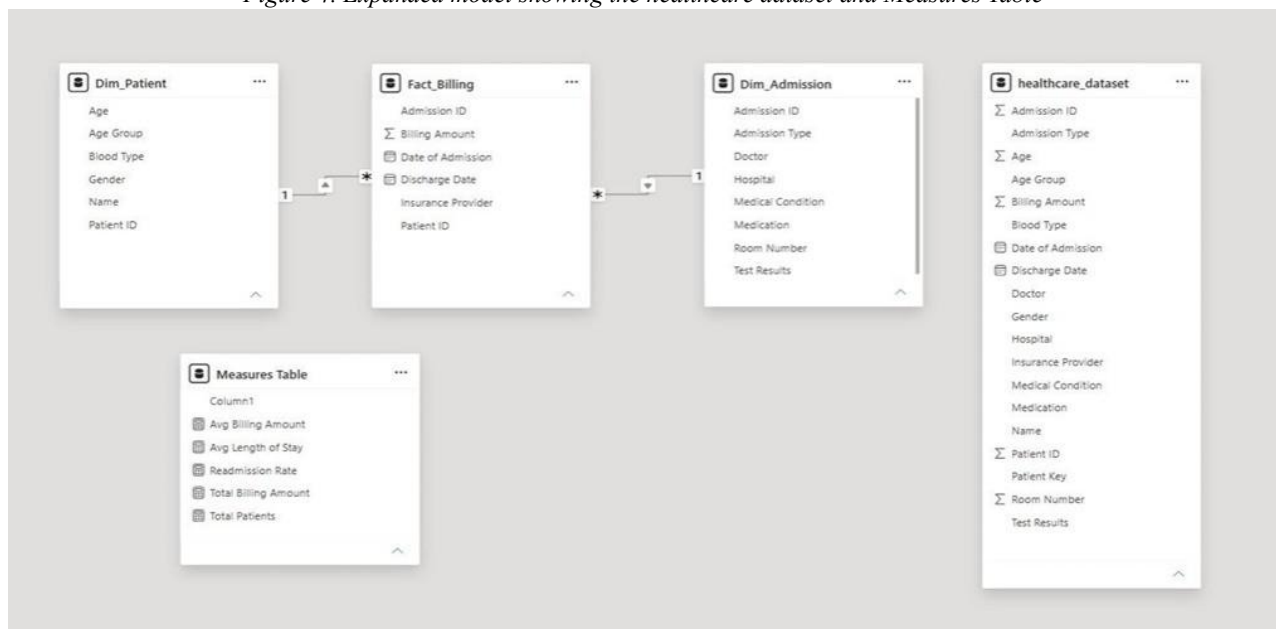


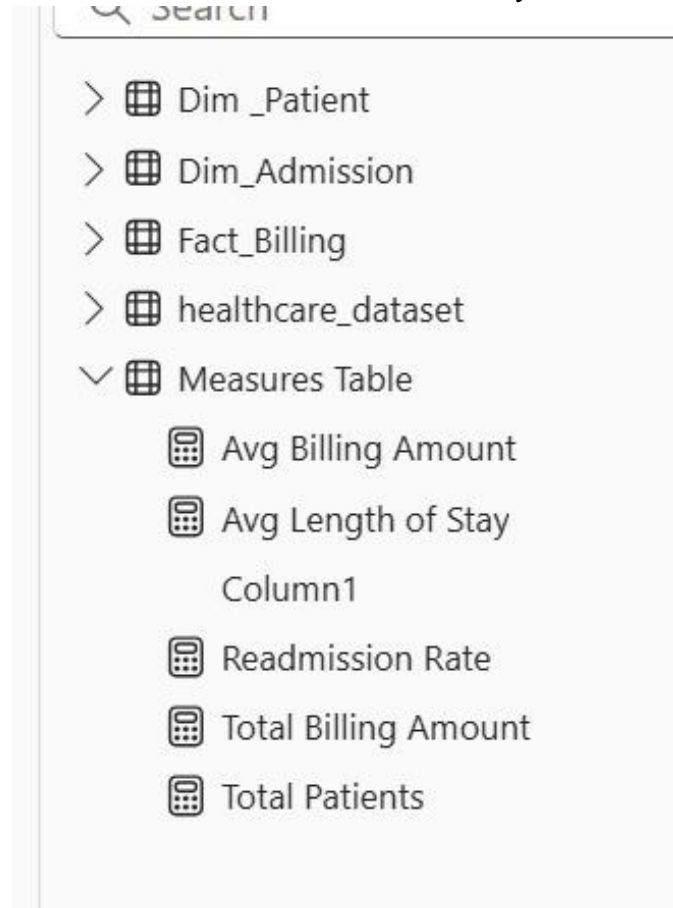
Figure 4. Expanded model showing the healthcare dataset and Measures Table



6. Measures Table

A separate Measures Table was created to store important DAX measures in one place. The visible measures include Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount and Total Patients. Keeping measures together makes the model easier to maintain and helps organize calculations for future report visuals.

Figure 5. Measures Table with analytical measures



7. Measure Creation

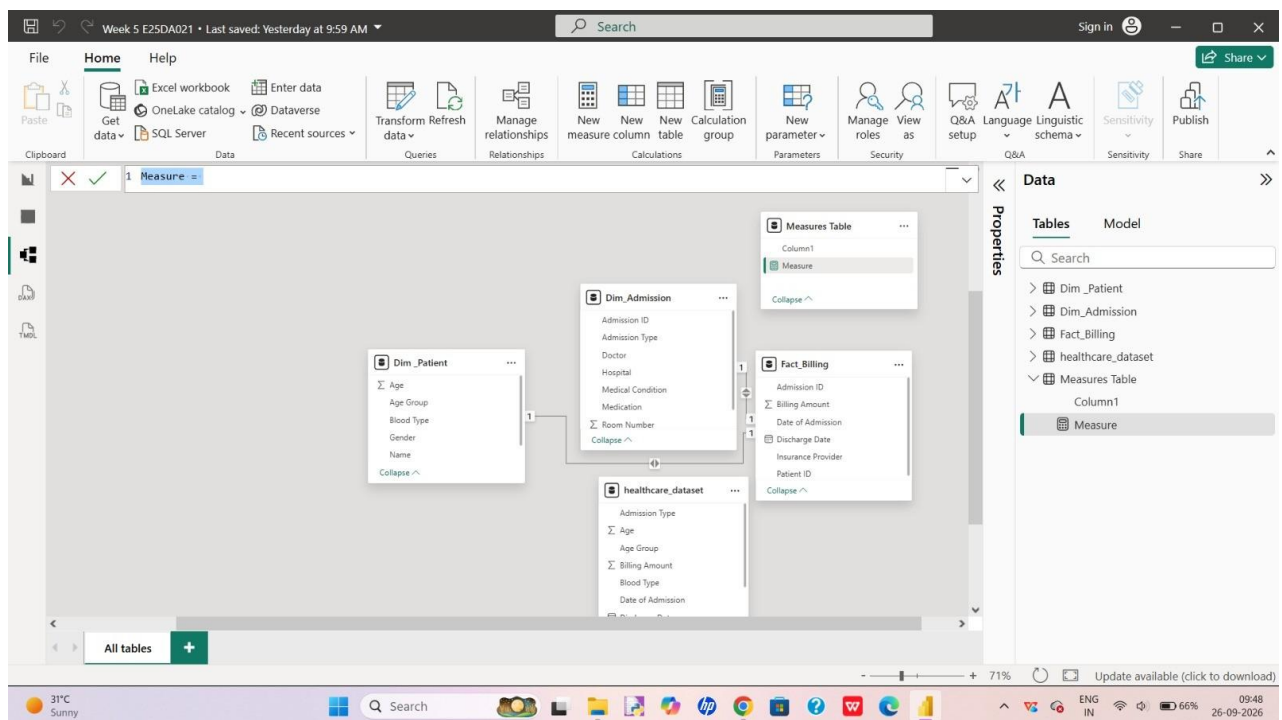
The screenshot also shows the creation of a DAX measure in the Measures Table. One visible calculation is Avg Billing Amount, which uses the Billing Amount field from Fact_Billing to calculate the average billing value. Similar measures are maintained in the same table for patient count, total billing, length of stay and readmission analysis.

Figure 6. Creating a measure in Power BI

8. Transformation Steps Performed

Step	Work Performed
Source	Loaded the healthcare dataset into Power Query.
Duplicate Removal	Removed duplicate records while preparing the patient dimension.
Column Selection	Removed unnecessary columns to keep the dimension table focused.
Index Creation	Added an index to generate a unique Patient ID.
Column Rename	Renamed the generated index column as Patient ID.
Merge	Merged billing and admission information using matching fields.
Data Model	Created relationships between fact and dimension tables.
Measures	Created a dedicated Measures Table for analytical calculations.

Figure 7. Power BI transformation steps in Dim_Patient



9. Result / Outcome

The Week 5 activity resulted in a more organized healthcare data model. The patient information was structured into a dimension table, billing information was maintained as a fact table, and admission details were organized separately. The merge operation connected related records, while relationships and measures prepared the model for interactive analysis.

The completed model can be used to build dashboards for studying billing amounts, patient counts, length of stay and readmission-related information. The Measures Table also provides a centralized location for analytical calculations.

10. Conclusion:

Week 5, Power BI was used not only for data visualization but also for data preparation and modeling. The practical work demonstrated Power Query transformations, table merging, relationship creation and DAX measure organization. These steps form a strong foundation for developing a meaningful healthcare analytics dashboard in the next stage.